# Changho Shin

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## ACADEMIC POSITIONS

**Princeton University**, Princeton, NJ — Feb. 2026 – Present
*Postdoctoral Researcher*
• Advisor: Brenden Lake

## EDUCATION

**University of Wisconsin–Madison** — Sep. 2020 – Dec. 2025
• Ph.D. in Computer Science; M.S. in Mathematics
• Advisor: Frederic Sala
• Thesis: *Learning from Weak Signals: Data-Centric Methods for Foundation Models*

**Seoul National University** — Mar. 2015 – Feb. 2017
• M.S. in Digital Contents and Information Studies
• Advisor: Professor Wonjong Rhee

**Seoul National University** — Mar. 2011 – Feb. 2015
• B.A. in Psychology, B.S. in Computer Science and Engineering
• Graduated with honors (Cum Laude)

## HONORS & AWARDS

**Qualcomm Innovation Fellowship Finalist** 2024
**Best Paper Award Honorable Mention**, NeurIPS R0-FoMo Workshop 2023
**NeurIPS Scholar Award** 2023
**1st Place**, DataComp Competition (Small Track, Filtering) 2023
**CS Departmental Scholarship**, University of Wisconsin-Madison 2020

## CONFERENCE PUBLICATIONS

[C10] Gabriel Orlanski, Devjeet Roy, Alexander Yun, **Changho Shin**, Alex Gu, Albert Ge, Dyah Adila, Nicholas Roberts, Frederic Sala, Aws Albarghouthi, "SlopCodeBench: Benchmarking How Coding Agents Degrade Over Long-Horizon Iterative Tasks", *Neural Information Processing Systems (NeurIPS)*, 2026.

[C9] **Changho Shin**, David Alvarez-Melis, "Curriculum Learning as Transport: Understanding Curricula with Wasserstein Geodesics", *Conference on Language Modeling (COLM)*, 2026.

[C8] Jitian Zhao*, **Changho Shin***, Tzu-Heng Huang, Srinath Namburi, Frederic Sala, "CARE: Confounder-Aware Aggregation for Reliable LLM Evaluation", *International Conference on Machine Learning (ICML)*, 2026.

[C7] **Changho Shin**, John Cooper, Frederic Sala, "Weak-to-Strong Generalization Through the Data-Centric Lens", *International Conference on Learning Representations (ICLR)*, 2025.

[C6] Yijing Zhang, Dyah Adila, **Changho Shin**, Frederic Sala, "Personalize Your LLM: Fake it then Align it", *North American Chapter of the Association for Computational Linguistics (NAACL) Findings*, 2025.

[C5] **Changho Shin**, Jitian Zhao, Sonia Cromp, Harit Vishwakarma, Frederic Sala, "OTTER: Improving Zero-Shot Classification via Optimal Transport", *Neural Information Processing Systems (NeurIPS)*, 2024.

[C4] Dyah Adila*, **Changho Shin***, Linrong Cai, Frederic Sala, "Zero-Shot Robustification of Zero-Shot Models", *International Conference on Learning Representations (ICLR)*, 2024.

**Best Paper Award Honorable Mention, Oral Presentation** at *NeurIPS 2023 R0-FoMo Workshop.*

[C3] **Changho Shin**, Sonia Cromp, Dyah Adila, Frederic Sala, "Mitigating Source Bias for Fairer Weak Supervision", *Neural Information Processing Systems (NeurIPS)*, 2023.

[C2] **Changho Shin**, Winfred Li, Harit Vishwakarma, Nicholas Roberts, Frederic Sala, "Universalizing Weak Supervision", *International Conference on Learning Representations (ICLR)*, 2022.

[C1] **Changho Shin**, Sunghwan Joo, Jaeryun Yim, Hyoseop Lee, Taesup Moon, Wonjong Rhee, "Subtask Gated Networks for Non-Intrusive Load Monitoring", *AAAI Conference on Artificial Intelligence*, 2019.

## JOURNAL PUBLICATIONS

[J2] **Changho Shin**, Eunjung Lee, Jeongyun Han, Jaeryun Yim, Hyoseop Lee, Wonjong Rhee, "The ENERTALK Dataset, 15 Hz Electricity Consumption Data from 22 Houses in Korea", *Scientific Data*, 2019.

[J1] **Changho Shin**, Seungeun Rho, Hyoseop Lee, Wonjong Rhee, "Data Requirements for Applying Machine Learning to Energy Disaggregation", *Energies*, May 2019.

## PREPRINTS

[P2] **Changho Shin**, Xinya Yan, Suenggwan Jo, Sungjun Cho, Shourjo Aditya Chaudhuri, Frederic Sala, "TARDIS: Mitigating Temporal Misalignment via Representation Steering", *arXiv*, 2025.

[P1] Amanda Dsouza, Christopher Glaze, **Changho Shin**, Frederic Sala, "Evaluating Language Model Context Windows: A 'Working Memory' Test and Inference-time Correction", *arXiv*, 2024.

## WORKSHOP PUBLICATIONS

[W7] **Changho Shin**, Jeonghwan Cheon, Brenden Lake, "Coarse-to-Fine Category Structure in Child's-View Model Representations", *NeurIPS 2026 DevAI Workshop*.

[W6] **Changho Shin**, Daiwei Chen, John Cooper, Brenden Lake, Frederic Sala, Ramya Korlakai Vinayak, "Compositional Self-Improvement", *ICML 2026 CompLearn Workshop*.

[W5] Sungjun Cho, **Changho Shin**, Suenggwan Jo, Xinya Yan, Shourjo Aditya Chaudhuri, Frederic Sala, "LLM-Integrated Bayesian State Space Models for Multimodal Time-Series Forecasting", *NeurIPS 2025 BERT2S Workshop*.

[W4] Dyah Adila, **Changho Shin**, Yijing Zhang, Frederic Sala, "Is Free Self-Alignment Possible?", *NeurIPS 2024 MINT Workshop*.

[W3] **Changho Shin***, Joon Suk Huh*, Elina Choi, "Pool-Search-Demonstrate: Improving Data-wrangling LLMs via better in-context examples", *NeurIPS 2023 TRL Workshop*. **Oral Presentation**.

[W2] **Changho Shin***, Tzu-Heng Huang*, Sui Jiet Tay, Dyah Adila, Frederic Sala, "Multimodal Data Curation via Object Detection and Filter Ensembles", *ICCV 2023 DataComp Workshop* (Rank #1 in DataComp competition filtering track (small)).

[W1] **Changho Shin**, Alice Schoenauer Sebag, "Can we get smarter than majority vote? Efficient use of individual rater's labels for content moderation", *NeurIPS 2022 ENLSP Workshop*.

## JOB EXPERIENCE

**Microsoft Research**, Cambridge — Jun. 2025 – Aug. 2025
*Research Intern*
• Mentor: David Alvarez-Melis
• Project: *Curriculum Learning as Transport: Understanding Curricula with Wasserstein Geodesics*

**Snorkel AI**, California — Jun. 2024 – Aug. 2024
*Research Intern*
• Mentor: Christopher Glaze, Paroma Varma
• Projects: *Reward Modeling, Synthetic Data Generation, LLM Evaluation*

**Twitter**, San Francisco — Jun. 2022 – Aug. 2022
*ML Engineer Intern*
- Mentor: Alice Schoenauer Sebag
- Manager: Milind Ganjoo
- *Improving toxicity classification via weak supervision [W1]*

**Encored Technologies**, Seoul, Korea — Jan. 2018 – Jul. 2020
*Data Scientist*
- Manager: Hyoseop Lee
- *Non-intrusive load monitoring [C1, J1, J2], Energy forecasting*

**Korea Institute for Defense Analyses**, Seoul, Korea — Jan. 2017 – Dec. 2017
*Researcher*

## TEACHING EXPERIENCE

**University of Wisconsin-Madison**
- Teaching assistant for CS 839 (Foundation Models) — Fall 2023, Fall 2025
- Teaching assistant for CS 300 (Programming II) — Fall 2022, Spring 2023
- Teaching assistant for CS 760 (Machine Learning) — Fall 2021, Spring 2022
- Teaching assistant for CS 320 (Data Programming II) — Spring 2021
- Teaching assistant for CS 220 (Data Programming I) — Fall 2020